

By defn, a pivot position of a matrix corresponds to a leading 1 of that row in reduced row echelon form.

$$\begin{aligned}
 B &= \begin{bmatrix} 3 & 1 & 2 \\ 2 & 5 & 6 \\ 5 & 6 & 8 \end{bmatrix} \xrightarrow[\text{+ Row 1}]{\text{Row 2 = Row 2}} \begin{bmatrix} 3 & 1 & 2 \\ 5 & 6 & 8 \\ 5 & 6 & 8 \end{bmatrix} \xrightarrow[\text{- Row 2}]{\text{Row 3 = Row 3}} \begin{bmatrix} 3 & 1 & 2 \\ 5 & 6 & 8 \\ 0 & 0 & 0 \end{bmatrix} \\
 &\xrightarrow[\text{(1/6)}]{\text{Row 2 = Row 2} \times} \begin{bmatrix} 3 & 1 & 2 \\ 5/6 & 1 & 4/3 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{- Row 2}]{\text{Row 1 = Row 1}} \begin{bmatrix} 13/6 & 0 & 2/3 \\ 5/6 & 1 & 4/3 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{x 6/13}]{\text{Row 1 = Row 1}} \begin{bmatrix} 1 & 0 & 4/13 \\ 5/6 & 1 & 4/3 \\ 0 & 0 & 0 \end{bmatrix} \\
 &\xrightarrow[\text{- 5/6 x Row 1}]{\text{Row 2 = Row 2}} \begin{bmatrix} 1 & 0 & 4/13 \\ 0 & 1 & 14/13 \\ 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

Verification of whether the final matrix is in reduced row echelon form

1. Row 1 and 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all non-zero rows must be above the zero rows is True.

2. Row 1 Leading element = 1 in Column 1

Row 2 Leading element = 1 in Column 2

$\therefore$ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0

Row 3 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry, all the positions below the leading entry must contain 0 is True.

4. Each leading entry is a 1, that condition is satisfied.

5. Row 1 Column 2 = 0

$\therefore$ The condition that for each column having a leading entry, all the posns. above the leading entry must contain 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 & 4/13 \\ 0 & 1 & 14/13 \\ 0 & 0 & 0 \end{bmatrix}$ is in reduced row echelon form. $\square$